

Sinusoidal steady-state analysis

ELEC0447 - Analysis of electric power and energy systems

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September 7, 2026

Overview

1. Basics and conventions
2. Sinusoidal steady-state analysis

What will we learn today?

Mostly from Chapter 2 of Ned Mohan's book:

Mohan, Ned. *Electric power systems: a first course*. John Wiley & Sons, 2012.

- ▶ Recap of conventions
- ▶ Sinusoidal steady-state analysis

You will be able to do exercises 2.1, 2.2, 2.4, 2.5, 2.9, 2.11, 2.12, 2.14, 2.16, 2.17, 2.18, 2.19 and 2.20 from Ned Mohan's book.

Basics and conventions

The background of the slide features a white central area where the text is located. This area is framed by two large teal-colored triangles that point towards each other from the left and right sides, meeting at a point at the bottom center. The overall effect is a clean, modern, and minimalist design.

Power and energy

- ▶ Power measures the rate of use of energy.
- ▶ It is expressed in Watt [W]: $1\text{ W} = 1\text{ Joule/second}$.
- ▶ In an electric system, the instantaneous power is:

$$p(t) = u(t) \times i(t)$$

- ▶ $u(t)$ is the voltage in volt [V] (line integral of electric field between two points).
 - ▶ $i(t)$ is the current in amperes [A].
 - ▶ t is time.
- ▶ To measure energy in power systems, we use units ranging from a kWh (a microgrid) to a TWh (a country).
- ▶ Devices have power ratings ranging from W to GW (although we generally speak in VA for ratings).

Power in perspective: The cyclist vs. the toaster

Can an Olympic cyclist generate enough electrical power to toast a single slice of bread?

The Cyclist vs. The Toaster

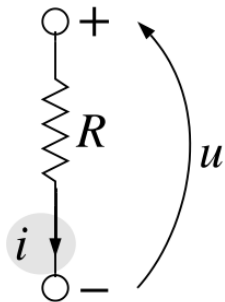
Robert Förstemann vs. 700W Toaster

Watch video: <https://www.youtube.com/watch?v=S405vo0CqAQ>

It takes an Olympic sprint cyclist generating nearly 700 W of mechanical power at maximum exertion for over a minute just to toast one single slice of bread!

Motor convention (or standard reference)

When using the **motor convention** to direct u w.r.t. i , $p(t)$ represents the power **consumed** by a device (here a resistor):



- ▶ The power consumed can be < 0 , $= 0$, or > 0 depending on the device.
- ▶ E.g., for a passive resistor we always have $p(t) \geq 0$.
- ▶ The **opposite** convention is the **generator convention** ($p(t)$ is the power produced).
- ▶ We will sometimes use a mix of both conventions, based on intuition, so that in general we have few negative numbers: **pay attention to the orientations!**

Magnetic fields, etc.

Magnetic fields play a central role in modeling the behavior of power system equipment (lines, transformers, generators, etc.).

As this course will not focus on detailed internal device physics, but rather on a *system of devices*, magnetic effects will often be highly abstracted (e.g., as an inductance, or a turns ratio).

Let's recall that:

- ▶ A magnetic field is due to charges in motion or magnetized materials.
- ▶ It is measured in A/m (for magnetic field strength H) or in Tesla [T] (for magnetic flux density $B = \mu H$, with μ_0 in vacuum/air).
- ▶ The magnetic flux (ϕ in Weber [Wb]) measures the magnetic field crossing a surface.
- ▶ A time-varying magnetic flux induces a voltage (Faraday-Lenz Law: $e(t) = -\frac{d\phi}{dt}$),

Sinusoidal steady-state analysis

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Sinusoidal signals and phasor representation

Unless otherwise specified, we will always work with sinusoidal signals and in steady state:

$$y(t) = \sqrt{2}Y \cos(\omega t + \phi_y)$$

- ▶ Y is the **rms** value of the signal.
- ▶ ϕ_y is its **phase angle**.
- ▶ $\omega = 2\pi f$ is its **angular frequency** in rad/s (f is the frequency in Hz).

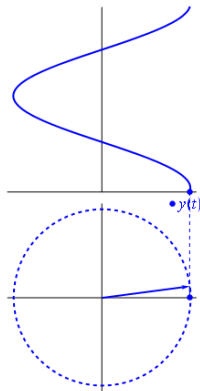
At a specific frequency $f = \frac{\omega}{2\pi}$, the signal can be represented as a **phasor**:

$$\bar{Y} = Y \angle \phi_y = Y e^{j\phi_y}$$

Phasors allow working in the frequency domain, converting differential equations into algebraic operations:

$$\mathcal{D} \left[\sqrt{2} \bar{Y} e^{j\omega t} \right] = \dots$$

Phasor concept



- ▶ A phasor represents a sinusoidal waveform rotating counter-clockwise at angular speed ω .
- ▶ Its projection on the real axis corresponds to the instantaneous value $y(t)$:

$$y(t) = \mathcal{Re}\{\sqrt{2}Y e^{j(\omega t + \phi_y)}\}$$

- ▶ Factoring out $e^{j\omega t}$ allows us to perform additions, subtractions, and circuit laws purely using complex algebra.
- ▶ Reference:

<https://en.wikipedia.org/wiki/Phasor>

Impedance

Let $u(t)$ and $i(t)$ be the voltage and current across a one-port, respectively, in sinusoidal steady state and with the motor convention:

- ▶ For a resistor: $u(t) = Ri(t) \implies \bar{U} = R\bar{I}$
- ▶ For an inductor: $u(t) = L \frac{di(t)}{dt} \implies \bar{U} = j\omega L \bar{I}$
- ▶ For a capacitor: $i(t) = C \frac{du(t)}{dt} \implies \bar{I} = j\omega C \bar{U} \implies \bar{U} = -j \frac{1}{\omega C} \bar{I}$

The **impedance** Z , a complex quantity, generalizes Ohm's law:

$$Z = R + jX \quad [\Omega], \quad \bar{U} = Z\bar{I}$$

- ▶ Resistor: $Z = R$ ($X = 0$)
- ▶ Inductor: $Z = jX = j\omega L$ ($X = \omega L > 0$)
- ▶ Capacitor: $Z = jX = -j \frac{1}{\omega C}$ ($X = -\frac{1}{\omega C} < 0$)

Impedance, admittance, etc.

The imaginary part of the impedance, X , is called **reactance**:

- ▶ $X > 0$: inductive reactance
- ▶ $X < 0$: capacitive reactance

The **admittance** Y is the inverse of the impedance, expressed in Siemens [S]:

$$Y = \frac{1}{Z} = G + jB$$

- ▶ G is the **conductance**
- ▶ B is the **susceptance**

Explicitly:

$$Y = \frac{1}{Z} = \frac{R - jX}{R^2 + X^2} = \frac{R}{R^2 + X^2} - j\frac{X}{R^2 + X^2}$$

Complex calculus

For an impedance $Z = R + jX$:

$$|Z| = \sqrt{R^2 + X^2}, \quad \angle Z = \arctan\left(\frac{X}{R}\right)$$

$$Z = \frac{\bar{U}}{\bar{I}} = \frac{U \angle \phi_u}{I \angle \phi_i} = \frac{U}{I} \angle (\phi_u - \phi_i)$$

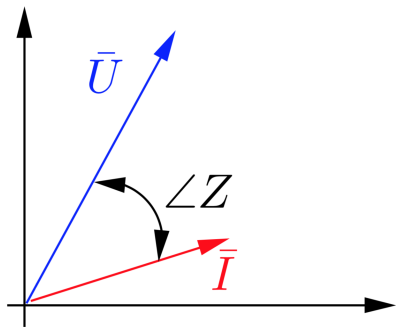
Animation

AC Circuit & Phasor Analysis Animation:

<https://www.youtube.com/watch?v=z07RZZW0wSQ>

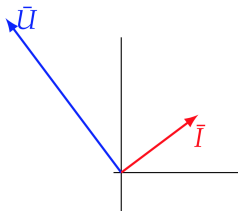
Phasor diagrams

Plotting the phasors in the complex plane provides immediate geometric insight:



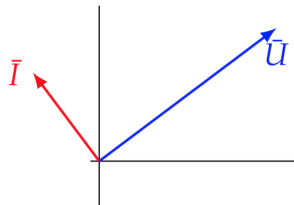
- ▶ One phasor (often the bus voltage $\bar{U} = U \angle 0^\circ$) is chosen as the reference along the positive real axis.
- ▶ The current phasor $\bar{I} = I \angle \phi_i$ is placed at angle ϕ_i .
- ▶ The phase difference is $\phi = \phi_u - \phi_i$.
- ▶ Angles are measured counter-clockwise (positive).

Inductive or capacitive? Which is which?



Inductive circuit ($X > 0$)

- ▶ Current $\bar{I}$ is **lagging** behind voltage $\bar{U}$
- ▶ $\phi_u > \phi_i \implies \phi = \phi_u - \phi_i > 0$
- ▶ Consumes reactive power ($Q > 0$)



Capacitive circuit ($X < 0$)

- ▶ Current $\bar{I}$ is **leading** ahead of voltage $\bar{U}$
- ▶ $\phi_u < \phi_i \implies \phi = \phi_u - \phi_i < 0$
- ▶ Supplies reactive power ($Q < 0$ in motor conv.)

The notions of power

The **complex power** is defined as:

$$S = \bar{U} \bar{I}^*$$

Let $\phi = \phi_u - \phi_i$, then:

$$S = (Ue^{j\phi_u})(Ie^{-j\phi_i}) = UIe^{j\phi} = P + jQ$$

- ▶ $P = UI \cos \phi$ is the **active power**, measured in Watt [W].
- ▶ $Q = UI \sin \phi$ is the **reactive power**, measured in volt-ampere reactive [var].
- ▶ $\cos \phi$ is the **power factor** (pf).
- ▶ The **apparent power** is $|S| = UI = \sqrt{P^2 + Q^2}$, measured in volt-ampere [VA].

Reactive power is, in general, undesirable: it increases the current magnitude $I = |S|/U$ for a given active power P , resulting in higher line losses (RI^2) and larger voltage drops.

Useful formulas: leading and lagging circuits

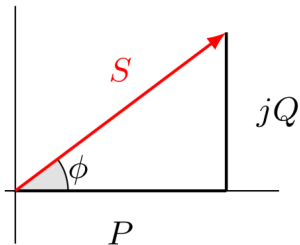
$$P = RI^2 = \frac{U_R^2}{R}, \quad Q = XI^2 = \frac{U_X^2}{X}$$
$$\tan \phi = \frac{Q}{P}, \quad \cos \phi = \frac{P}{|S|}$$

Leading vs. Lagging

The value of the power factor $\cos \phi \in [0, 1]$ alone does not indicate the sign of ϕ :

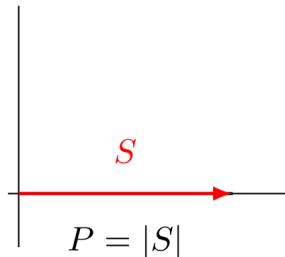
- ▶ In an **inductive** system, $u(t)$ precedes $i(t)$, $i(t)$ is **lagging**, thus $Q > 0$ (motor convention).
- ▶ In a **capacitive** system, the opposite holds: $i(t)$ is **leading**, thus $Q < 0$.

Power factor compensation



Uncompensated load draws active power P and reactive power $Q_L > 0$. Total line current:

$$I_1 = \frac{\sqrt{P^2 + Q_L^2}}{U}.$$



Connecting a parallel capacitor provides $Q_C = -\omega CU^2 < 0$. Total reactive power is canceled: $Q_{net} \approx 0$. Total current drops to $I_2 \approx \frac{P}{U} < I_1$!

Power factor compensation: Example

Problem statement

A 120 V (rms) voltage source at 60 Hz feeds an R-L load:

$$S_{load} = 1858.4 + j1031.4 \text{ VA}$$

- ▶ Active power: $P = 1858.4 \text{ W}$
- ▶ Reactive power: $Q = 1031.4 \text{ var}$ (inductive, lagging)
- ▶ Apparent power: $|S| = \sqrt{P^2 + Q^2} = 2125 \text{ VA}$
- ▶ Initial power factor: $\cos \phi = \frac{P}{|S|} = \frac{1858.4}{2125} \approx 0.875$ lagging
- ▶ Initial line current: $I = \frac{|S|}{U} = \frac{2125}{120} \approx 17.7 \text{ A}$

Power factor compensation: Example (cont'd)

Goal: Compensate to unity power factor ($\cos \phi_{new} = 1.0$, $Q_{new} = 0$).

The parallel capacitor must supply reactive power equal and opposite to Q_{load} :

$$Q_C = -Q_{load} = -1031.4 \text{ var} = -\omega C U^2$$

Solving for C :

$$C = \frac{Q_{load}}{\omega U^2} = \frac{1031.4}{2\pi \times 60 \times 120^2} \approx 189.9 \mu\text{F}$$

Result on line current:

$$I_{new} = \frac{P}{U} = \frac{1858.4}{120} \approx 15.49 \text{ A}$$

This represents a **12.5% reduction in line current** with zero change in delivered active power P !

Summary

- ▶ **Conventions:** Motor convention defines consumed power $p(t) = u(t)i(t)$; generator convention defines produced power.
- ▶ **Phasors:** Represent sinusoidal quantities $y(t) = \sqrt{2}Y \cos(\omega t + \phi_y)$ as algebraic complex numbers $\bar{Y} = Y e^{j\phi_y}$.
- ▶ **Impedance & Admittance:** $Z = R + jX$ and $Y = G + jB = 1/Z$.
- ▶ **Complex Power:** $S = \bar{U}\bar{I}^* = P + jQ = |S|e^{j\phi}$.
 - ▶ $P = UI \cos \phi$ (active power, in Watt [W])
 - ▶ $Q = UI \sin \phi$ (reactive power, in var [var])
 - ▶ $|S| = UI$ (apparent power, in Volt-Ampere [VA])
- ▶ **Power Factor:** $\cos \phi = P/|S|$; lagging for inductive loads, leading for capacitive loads.
- ▶ **Power Factor Correction:** Shunt capacitors supply reactive power locally, reducing current flows, thermal losses, and voltage drops across the network.